

SAFA TAHA

Tbilisi, Georgia

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Education

Bachelor of Science in Electrical and Electronic Engineering

Ilia State University

Expected July 2027

Relevant Experience

April 2023 – March 2025

Program Ambassador

Google's Women Techmaker's Program MENA

- Co-organized and mentored in a week-long hands-on "IoT using ESP32" workshop with 50+ participants, focusing on microcontroller architecture, sensor interfacing, low-power wireless communication protocols, and real-world IoT projects.
- Guided participants in embedded project development, including sensor integration and real-time data acquisition.
- Mentored 10 Artificial Intelligence and Machine Learning projects guiding participants across domains such as Retrieval-Augmented Generation (RAG) systems, NLP, supervised and unsupervised learning.
- Co-developed a Python programming bootcamp curriculum with emphasis on engineering applications and system-level thinking, creating project-based modules and teaching foundational programming concepts such as OOP, data structures, and API integration.

July 2024 – October 2024

Embedded AI Engineering Intern

ElectroPI, Cairo, Egypt

- Designed and deployed on-device and edge-assisted AI pipelines, integrating RAG architectures (LangChain + ChromaDB) with ESP32 communication protocols (Wi-Fi/Bluetooth) to enable real-time bilingual assistant functionality with ~90% retrieval accuracy.
- Developed and optimized ESP32 based embedded systems for edge AI applications, reducing execution latency by 67% through efficient firmware design, and task scheduling.
- Applied LSTM, ANN, and CNN models for predictive maintenance and financial analysis, focusing on low latency data acquisition and processing pipelines on ESP32, interfacing with sensors and external APIs to support real-time inference and decision-making.

Professional Profile

Key Projects

- Multi MCU IoT Development Board (STM32F4 + ESP32-S3) - 2026**
 - IoT board with inter-MCU communication interfaces (UART/SPI), optimized signal integrity, mixed RF and digital signals, power management circuitry including voltage regulation stages, decoupling strategies, and clock systems for stable operation.
- ISL6236A Quad-Output Voltage Supply PCB - 2026**
 - High efficiency power supply module based on the ISL6236A controller with feedback networks and inductor sizing to provide stable voltage rails for sensitive electronic components.
- STM32 Self-Balancing Two Wheeled Robot - 2026**
 - PID control system using an MPU6050 IMU to calculate pitch and roll via a Kalman Filter, utilizing control loops to drive DC motors.

Skills

- Embedded Systems & Hardware:** STM32 (F0/F4/CubeIDE), ESP32 (S3/Mini/WROOM), Raspberry Pi, Arduino, Verilog, Altium, KiCad, LTspice, Proteus
- Protocols:** I2C, SPI, UART, CAN, PWM, DMA, ADC, DAC
- Data & Modeling:** Pandas, NumPy, Matplotlib, Seaborn, MATLAB, Octave, Fusion 360
- Tools:** VS Code, Jupyter, Logisim, Git